Physical Model of the Coupled Levitated-Sensor Array: Equations and Optimization Results

George Issa

August 31, 2026

Abstract

This note documents the physical (SI) sensitivity pipeline of `ResOSc` after anchoring it to the published LSD parameters [PRL **128**, 111101 (2022), Table I] and to the levitated-sensor proposal of Arvanitaki & Geraci [PRL **110**, 071105 (2013), "AG13"]. Every equation used in the pipeline is stated with all terms defined (Sec. 1); parameters are tabulated (Sec. 2); the Metropolis optimization and the three-way comparison results are summarized (Secs. 3–4). The main result can be described as follows: the cavity readout (A) is thermal-noise-limited, and its optimal array is the uniform stack with maximum trap (three-way tie, $d_{\max} = 7.8$ pc at $\mathcal{M}_c = 10^{-3} M_\odot$). However, the imaging readout (B) is shot-noise-limited. There, *Coulomb coupling wins*: the optimized coupled array reaches 2.2× the best uncoupled array and 12× readout B's own as-built baseline, with its deepest sensitivity buckets on the thermal floor inside the LSD target band. Absolute forecasts currently hinge on a small set of unconfirmed parameters and conventions collected as questions in Sec. 5.

1 The sensitivity chain, with every term defined

We consider N optically levitated discs in a cavity of length L, with positions $x_i(t)$ measured along the cavity axis.

1.1 Mechanics and normal modes

The equations of motion are

$$M\ddot{\mathbf{x}} + (\text{damping}) + K\mathbf{x} = \mathbf{F}(t), \tag{1}$$

where $\mathbf{x} = (x_1, \ldots, x_N)^T$ are the axial displacements [m], $M = \mathrm{diag}(m_1, \ldots, m_N)$ the masses [kg], $\mathbf{F}(t)$ the external forces [N], and K [N/m] the stiffness matrix assembled from the optical trap stiffnesses $k_{\mathrm{trap},i} = m_i(2\pi f_{\mathrm{trap},i})^2$ ($f_{\mathrm{trap},i}$ = trap frequency [Hz]) on the diagonal plus the inter-particle couplings k_{ij} in the standard spring (graph-Laplacian) pattern: $K_{ii} = k_{\mathrm{trap},i} + \sum_{j\neq i} k_{ij}$, $K_{ij} = -k_{ij}$. The couplings come from Coulomb interaction between charges q_i (units of the elementary charge e) at equilibrium separations d_{ij}:

$$k_{ij} = \frac{2\, q_i q_j e^2}{4\pi\varepsilon_0\, d_{ij}^3}, \tag{2}$$

with ε_0 the vacuum permittivity; all pairs are included ($1/d^3$ tails), and equal-sign charges give $k_{ij} > 0$ (repulsive longitudinal spring). Adjacent-disc spacing: $d = 15.5\,\mu$m (companion SNR note).

The generalized eigenproblem $K\mathbf{v} = \omega^2 M\mathbf{v}$ yields normal modes $n = 1..N$ with frequencies $f_n = \omega_n/2\pi$ [Hz] and unit-norm shapes $\mathbf{v}_n$ (columns of V). Modal masses and the two modal damping rates (see Sec. 1.3 for the distinction) are mass-weighted averages over the shape:

$$\mu_n = \mathbf{v}_n^T M \mathbf{v}_n \quad [\text{kg}], \qquad \Gamma_n = \frac{\sum_i V_{in}^2 m_i \gamma_i}{\mu_n}, \qquad \Gamma_n^{\text{f}} = \frac{\sum_i V_{in}^2 m_i \gamma_i^{\text{f}}}{\mu_n} \quad [\text{Hz}], \tag{3}$$

where γ_i is the per-disc mechanical *linewidth* and γ_i^{f} the per-disc *force-noise* damping, both in Hz (cycles/s; angular rates are $2\pi\times$ these). Each mode responds to force at frequency f through the susceptibility

$$\chi_n(f) = \frac{1}{\mu_n[(2\pi f_n)^2 - (2\pi f)^2 + i\,(2\pi f)(2\pi\Gamma_n)]} \quad [\text{m/N}]. \tag{4}$$

1.2 Gravitational-wave drive (AG13, exact)

A GW of strain $h(t)$ (dimensionless), propagating perpendicular to the cavity, drives disc i with the inertial force [AG13 Eq. (5)]

$$f_i^{\text{sig}}(t) = \tfrac{1}{2}\, m_i\, (2\pi f_{\text{gw}})^2\, (L - x_i)\, h(t) \;\approx\; \tfrac{1}{2}\, m_i\, (2\pi f_{\text{gw}})^2\, L\, h(t), \tag{5}$$

where f_{gw} is the *wave's* frequency, L the *full* cavity length, and $x_i \ll L$ (discs near the input mirror). Physically: the trap minima are locked to the intracavity's interference pattern, which breathes with the wave by $\frac{1}{2}hL$. The disc's inertia resists, giving a force = mass × reference acceleration. Equation (5) is exact at all frequencies and gives a free mass displacement response $\frac{1}{2}hL$ above resonance, much like the flat response seen in LIGO detectors. The companion note's Eq. (20), $\beta_i = k_{\text{trap},i}L$, is the quasi-static ($f_{\text{gw}} \to$ resonance) limit *with L the half-baseline*. It seems that two conventions differ by a factor 2 in L (Question 6).

The modal drive and the (strain to observable) transfer function are

$$B_n(f) = \tfrac{1}{2}(2\pi f)^2 L\, (\mathbf{v}_n^T \mathbf{m}) \quad [\text{N/strain}], \qquad T(f) = \sum_n N_n\, B_n(f)\, \chi_n(f) \quad [\text{m/strain}], \tag{6}$$

where $\mathbf{m} = (m_1, \ldots, m_N)^T$ and $N_n = \mathbf{v}_n^T \mathbf{w}$ is the modal projection of the readout weight vector $\mathbf{w}$ (the observable is the single time series $O(t) = \sum_i w_i x_i(t)$).

1.3 Thermal noise and the damping split

Two distinct damping rates enter the thermal noise (one of the most recent changes):

- **Force-noise damping** γ_i^{f}: real bath friction, obeying the fluctuation–dissipation theorem (FDT). It is expressed as the sum of residual gas damping and the equivalent of the photon recoil heating,

$$\gamma_i^{\text{f}} = \gamma_g + \frac{\hbar\,(2\pi f_{\text{trap},i})}{k_B T}\, \gamma_{\text{sc},i}, \qquad \gamma_g = \frac{\hbar\,(2\pi f_0)}{k_B T}\, (N_i \gamma_g)_{\text{pub}} \approx 2.7 \times 10^{-9}\ \text{Hz}, \tag{7}$$

  where T is the gas temperature [K], k_B Boltzmann's constant, $\hbar$ the reduced Planck constant, $\gamma_{\text{sc},i}$ the photon-recoil rate (0.05 Hz at $f_{\text{trap}} = 100$ kHz, scaling $\propto f_{\text{trap}}$; PRL 128 Table I), and $(N_i\gamma_g)_{\text{pub}} = 0.17$ Hz at 100 kHz is the published *product* of the mean phonon occupation $N_i = k_B T/\hbar\omega_0 \approx 6.3 \times 10^7$ and the gas damping. N.B. the published 0.17 Hz is *not* the mechanical linewidth (Question 1).

- **Mechanical linewidth** $\gamma_i = f_{\mathrm{trap},i}/Q_{\mathrm{eff},i}$: set by feedback cooling (cold damping), which adds damping *without* force noise in the idealized limit. $Q_{\mathrm{eff},i}$ is a control setting set for every disc (AG13 uses 5.4×10^5 and claims it is tunable).

The modal thermal force PSD (power spectral density) and its appearance at the observable are

$$S_{Q_n} = 4k_B T\,\mu_n\,(2\pi\Gamma_n^{\mathrm{f}}) \;\; [\mathrm{N}^2/\mathrm{Hz}], \qquad S_O^{\mathrm{th}}(f) = \sum_n N_n^2\,S_{Q_n}\,|\chi_n(f)|^2 \;\; [\mathrm{m}^2/\mathrm{Hz}]. \tag{8}$$

Because thermal noise is a force, it is filtered by the same $|\chi_n|^2$ as the signal. On resonance, the thermal limited strain noise is therefore independent of both linewidth and weight. When mode n dominates at $f = f_n$, Eqs. (6), (13) and (8) give

$$S_h\big|_{f_n} \;=\; \frac{S_O^{\mathrm{th}}}{|T|^2} \;=\; \frac{N_n^2\,S_{Q_n}\,|\chi_n|^2}{N_n^2\,B_n^2\,|\chi_n|^2} \;=\; \frac{S_{Q_n}}{B_n^2(f_n)}, \tag{9}$$

where the weights cancel through N_n^2 (they multiply signal and thermal noise identically) and the susceptibility (with it the mechanical linewidth Γ_n, hence Q_{eff}) cancels through $|\chi_n|^2$. What survives is force noise over drive; note $S_{Q_n} \propto \Gamma_n^{\mathrm{f}}$, the *force-noise* damping, so the thermal floor depends on the bath, never on the feedback. The intuitive picture that might be happening is that readout cannot distinguish a strain-driven wiggle from a thermally-driven one, since both are real motion through the same spring, so no reweighting or amplification changes their ratio.

1.4 Readout (shot) noise

Readout A (cavity probe; weights hardwired). All discs dispersively shift one cavity mode with a coupling per disc:

$$G = k_c\,\frac{V_d}{4V_c}\,(\epsilon - 1)\,\omega_c \;\; [\mathrm{rad\,s^{-1}m^{-1}}], \qquad V_c = \frac{\pi w_0^2 L}{4}, \tag{10}$$

with $k_c = 2\pi/\lambda$ the probe wavevector, $\omega_c = 2\pi c/\lambda$ its angular frequency, $\lambda = 1.55\,\mu$m, V_d the disc (stack) volume, ϵ its dielectric constant, w_0 the cavity waist, and c the speed of light. The measured phase carries $\sum_i G_i x_i$, fixing the weights $\mathbf{w} = \mathbf{G}/|\mathbf{G}|$. The phase shot noise $S_\phi = \hbar\omega_c/(\eta P_{\mathrm{det}})$ $[\mathrm{rad}^2/\mathrm{Hz}]$ (η = detection quantum efficiency, P_{det} = detected power) gives the displacement-equivalent readout noise

$$S_{O,A}^{\mathrm{ro}}(f) = \frac{S_\phi}{\sum_i G_i^2}\left(\frac{\kappa}{4}\right)^2\left[1 + \left(\frac{f}{f_{\mathrm{pole}}}\right)^2\right], \qquad \kappa = \frac{\pi c}{\mathcal{F}L}, \quad f_{\mathrm{pole}} = \frac{\kappa}{4\pi}, \tag{11}$$

with κ the cavity *energy* decay rate [rad/s] and $\mathcal{F}$ the cavity finesse. Equation (11) *is* AG13's impedance-matched reflection result, $\sqrt{S_z} = \frac{\kappa}{4k_c g}\sqrt{\hbar\omega_c/P_d}\,\sqrt{1 + 4\omega^2/\kappa^2}$ (their shot-limited phase sensitivity $\delta\phi \approx 1/(2\sqrt{I})$, $I = P_d/\hbar\omega_c$), which we adopted the signal sidebands are filtered by the cavity *field-amplitude* half-linewidth, so the pole sits at $\omega = \kappa/2$, i.e. $f_{\mathrm{pole}} = \kappa/4\pi$. (Earlier, we had used prefactor $\kappa/2$ and pole $\omega = \kappa$; the two agree far above the pole but the generic form is $4\times$ noisier in power below it). Now, disc parameters (100 m cavity, $\mathcal{F} = 10$, $P_d = 0.2$ mW), Eq. (11) evaluated at their 100 kHz operating point gives $4.0 \times 10^{-16}\,\mathrm{m}/\sqrt{\mathrm{Hz}}$ vs. their quoted 3.8×10^{-16}. For LSD ($L = 10$ m, $f_{\mathrm{pole}} = 0.75$ MHz) the whole 10–300 kHz band lies below the pole. Under the corrected (quieter) form, readout A's corner-system horizon rises $\approx \sqrt{2}$, from 1.14×10^6 to 1.61×10^6 AU (7.8 pc).

Readout B (per-disc imaging; weights free). Each disc is imaged independently and carries a position imprecision

$$S_i^{\rm ro} = \frac{\hbar\,\omega_{\rm ro}\,\lambda_{\rm ro}^2}{4\pi\,\eta\,{\rm NA}^2\,P_i^{\rm sc}}\ \ [{\rm m}^2/{\rm Hz}], \qquad S_{O,B}^{\rm ro} = \sum_i w_i^2\,S_i^{\rm ro}, \tag{12}$$

where $\lambda_{\rm ro}$ ($\omega_{\rm ro} = 2\pi c/\lambda_{\rm ro}$) is the imaging wavelength, NA the collection numerical aperture, and $P_i^{\rm sc}$ the collected scattered power. The weights $\mathbf{w}$ are a free (unit norm) post-processing choice. Therefore, weights that depend on frequency, $\mathbf{w}(f)$, are a strict further improvement not yet implemented.

1.5 Strain-referred noise, source, and figure of merit

$$S_h(f) = \frac{S_O^{\rm th}(f) + S_O^{\rm ro}(f)}{|T(f)|^2}\ \ [{\rm Hz}^{-1}] \tag{13}$$

is the strain-referred noise PSD. Its square root, $\sqrt{S_h}$, is the plotted sensitivity curve. The source is an equal mass primordial black hole binary of chirp mass $\mathcal{M}_c = (m_1 m_2)^{3/5}/(m_1+m_2)^{1/5}$ with the stationary phase inspiral spectrum (Maggiore)

$$|\tilde{h}(f)|^2 = \mathcal{A}^2 f^{-7/3}, \qquad \mathcal{A}^2 = \frac{5}{24\,\pi^{4/3}}\frac{c^2}{r^2}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/3}\Theta, \tag{14}$$

with r the source distance, G Newton's constant, and Θ an $O(1)$ orientation factor (set to 1). The spectrum ends at the innermost stable circular orbit frequency $f_{\rm ISCO} = c^3/(6^{3/2}\pi G M_{\rm tot})$ with $M_{\rm tot} = 2^{6/5}\mathcal{M}_c$ for equal masses (≈ 1.9 MHz at $\mathcal{M}_c = 10^{-3}M_\odot$: the entire 10–300 kHz band is live). The matched-filter signal-to-noise ratio and horizon distance are

$$\rho^2 = 4\int_{f_{\rm lo}}^{f_{\rm hi}} \frac{|\tilde{h}(f)|^2}{S_h(f)}\,df, \qquad d_{\rm max} = r_{\rm ref}\,\frac{\rho(r_{\rm ref})}{\rho_{\rm thr}}, \quad \rho_{\rm thr} = 8, \tag{15}$$

using $\rho \propto 1/r$. Band: 10–300 kHz; benchmark $\mathcal{M}_c = 10^{-3}M_\odot$ ($\approx$ one Jupiter mass; middle of the PRL 133, 111401 search range).

2 Parameters and validation

Validation. The pipeline reproduces the published on-resonance strain sensitivities $h_{\rm min} = 1.02\times 10^{-22}$ (100 kHz) and 7.6×10^{-21} (10 kHz) at ratios 3.95 and 4.66 respectively — a *consistent* systematic offset compatible with the published damping table being angular rates ($\sqrt{2\pi}\approx 2.5$) times $O(1)$ bookkeeping; the 10-vs-100 kHz *scaling* agrees to 18% (Question 1). Four further internal validation tests (single-oscillator FDT peak, weight-independence of thermal S_h, per-mode SNR vs. full integral, integration-grid convergence to 10^{-4} of brute force) all pass.

3 Optimization setup

Annealed Metropolis walk minimizing $E = -\log d_{\rm max}$: log-space multiplicative proposals on positive variables (clipped to bounds), tangent-space Gaussian on $\mathbf{w}$. Geommetric annealing T : $0.2 \to 0.002$ (in log-$d_{\rm max}$ units); 10^5 steps per chain; three chains per case (one seeded at the corner, two random). The reported result is best ever visited state. Design variables and bounds:

Table 1: Benchmark parameters and provenance. "128" = PRL 128, 111101 Table I; "AG13" = PRL 110, 071105; "note" = companion SNR note; **TBC** = unconfirmed.

Parameter	Symbol	Value	Source
Trap frequency range	f_{trap}	10–100 kHz	128
Cavity length	L	10 m	128
Pressure / temperature	P, T	10^{-11} Torr, 300 K	128
Occupation×gas damping	$N_i\gamma_g$	0.17 Hz (100 kHz)	128 (Q1)
Photon-recoil rate	γ_{sc}	0.05 Hz (100 kHz), $\propto f$	128 (Q1)
Derived gas damping	γ_g	2.7×10^{-9} Hz	derived
Effective Q (feedback)	Q_{eff}	5.4×10^5 baseline; cap **TBC**	AG13 (Q2)
Disc stack mass	m	5.8×10^{-10} kg	geometry ✓
Stack volume, $\epsilon - 1$	V_d, $\epsilon - 1$	2.6×10^{-13} m^3, 1.1	geometry
Probe wavelength, waist	λ, w_0	1.55 μm, 75 μm	128/AG13
Cavity finesse	$\mathcal{F}$	10 ($f_{\text{cav}} = 1.5$ MHz)	AG13
Detection power (A)	P_{det}	0.2 mW	AG13
Imaging aperture (B)	NA	0.5 ("up to"; was 0.1)	Nancy 8/31 (Q3)
Imaging wavelength (B)	λ_{ro}	532 nm	note, **TBC**
Collected power (B)	P_i^{sc}	1 μW	**TBC** (Q3)
Quantum efficiency	η	0.8	**TBC** (Q3)
Charge range / stray floor	q	$[10^3, 10^8]\, e$	**TBC** (Q4)
Intracavity profile	G_i	uniform assumed	**TBC** (Q5)

$f_{\text{trap},i} \in [10, 100]$ kHz; $q_i \in [10^3, 10^8]\, e$ (coupled cases); $Q_{\text{eff},i} \in [10^3, 10^8]$ (upper bound = engineering constraint, Q2); $\mathbf{w}$ free for readout B only. Three-way comparison: (i) *corner system* baseline: uniform traps at 100 kHz, charges at the stray floor, $Q_{\text{eff}} = 5.4 \times 10^5$, uniform weights; (ii) *uncoupled opt*: charges removed, everything else optimized; (iii) *coupled opt*: everything optimized.

4 Results

Table 2: Three-way comparison, $d_{\max}$ at $\mathcal{M}_c = 10^{-3} M_\odot$ [AU]. v5: NA = 0.1, Q_{eff} fixed at 5.4×10^5. v7: NA = 0.5, per-disc Q_{eff} optimized, corrected readout-A shot-noise convention (readout B is identical to the earlier v6 run, which the correction does not touch).

	Readout A		Readout B	
Case	v5	v7	v5	v7
Corner system	1,138,752	1,605,008	83.9	419.7
Uncoupled opt	1,138,752	1,605,008	208.9	2,260.6
Coupled opt	1,138,752	1,605,008	380.9	**4,904.9**
coupled/uncoupled	1.000	1.000	1.82	**2.17**

Readout A (thermal-limited in its bucket): exact tie at $d_{\max} = 7.78$ pc. The optimum is the uniform 10 disc stack at the 100 kHz trap ceiling. Its bucket bottom (1.3×10^{-22}) reaches the LSD target band. Coupling is provably harmless there (any coupling network exerts zero net force on the common mode of identical traps) and empirically unhelpful. Q_{eff} is a null direction because

signal and thermal noise amplify identically.

Readout B (shot-limited): the hierarchy is strict and converged (seed spread 3–6%): corner < uncoupled opt < coupled opt. The v6 winner has traps in two clusters (20–32 and 74–100 kHz), charges (0.3–1) $\times 10^8\, e$, modes spread 63–383 kHz, with several of them *above* the trap ceiling, reachable only through coupling and fully driven under Eq. (5). Every Q_{eff} pinned at (0.6–1) $\times 10^8$: the deepest buckets sit *on the thermal floor* inside the LSD target band (Fig. 1). Because Q_{eff} pins at the search bound, readout B's absolute reach is *bound-limited*: it scales directly with the practical Q_{eff} cap (Question 2). The relative statements (coupled beats uncoupled 2.17×; both beat the corner) are insensitive to global factors.

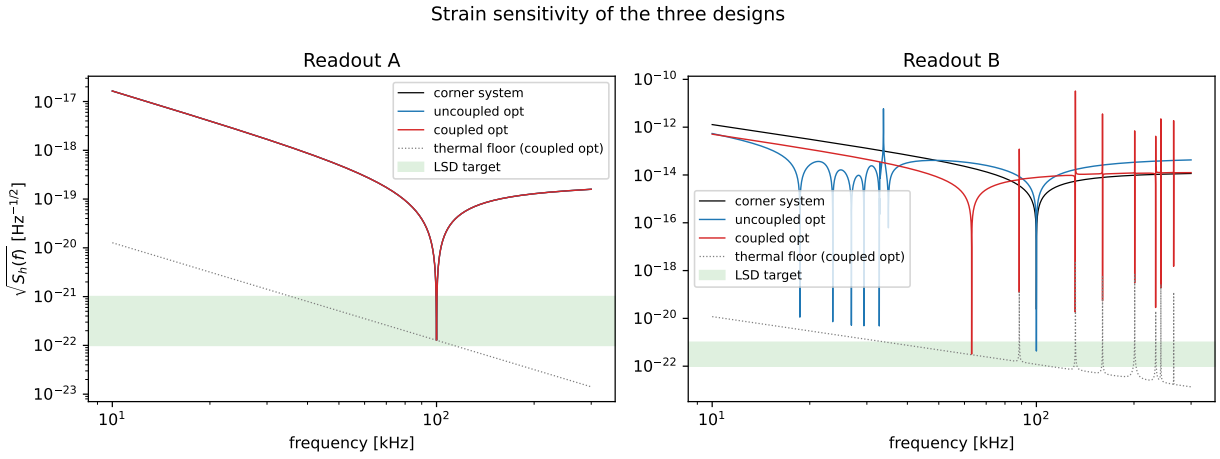


Figure 1: Strain sensitivity of the three designs (v7). Left: readout A (tie; single bucket at 100 kHz on its thermal floor). Right: readout B — the coupled optimum (red) tiles buckets across and beyond the trap range; its deepest dips touch the thermal floor (gray dotted) inside the LSD target band (green).

5 Questions and notes I am not sure of

Q1. Units of the damping table (factor 2π). In PRL 128 Table I, are $N_i\gamma_g$ (0.17/1.7 Hz) and γ_{sc} (0.05/0.005 Hz) plain frequencies (cycles/s) or angular rates (rad/s) labeled Hz? The FDT force noise depends on the angular rate, so the answer moves every absolute $\sqrt{S_h}$ and $d_{\max}$ by $\sqrt{2\pi} \approx 2.5$ (16× in event rate). I have tried to fit both, but I still can't figure it out: The table's own column headers are $\omega_0/2\pi = 10/100$ kHz, showing the authors distinguish angular from cycle frequencies and suggesting the damping rows have units Hz. But feeding the table into the paper's own Eq. (1) (whose $Q = \omega_0/\gamma_g$ requires the *angular* rate) reproduces the published $h_{\min} = 1.02 \times 10^{-22}$ to within 1.6× if the tabulated values are read as angular rates, but only to within 4.0× if read as plain Hz.

Q2. Practical ceiling on Q_{eff}. How high can the feedback cooling Q realistically go? In other words, how weak can cold damping be while (a) the trap frequency stays stable to within a linewidth over the $\sim 1/\Gamma$ coherence time (laser intensity noise), and (b) feedback measurement noise injection remains negligible? Our optimizer pins at the arbitrary search bound 10^8, and readout B scales with the true cap.

Q3. Readout B (imaging) parameters. Is NA = 0.5 an operating point or an upper limit, and is it simultaneously achievable for all ten discs? Collected power per disc P_i^{sc} (we assume 1 μW), imaging wavelength (532 nm?), and detector quantum efficiency η (0.8?). These set the shot plateau, Eq. (12), and hence all readout B numbers.

Q4. Charge range and stray floor. What charge can be controllably placed on a disc (upper bound), and what is the residual/stray charge floor (lower bound)? These bound the Coulomb design space, Eq. (2). The coupled optimum currently uses $q \sim 10^8\, e$. Is this reasonable on a 75 μm stack?

Q5. Intracavity coupling profile G_i. Are the ten discs' dispersive couplings equal (all at equivalent antinode positions), or is there a known non-uniform profile? A non-uniform hardwired profile is the one regime where coupling could also matter for readout A (mechanically re-aligning modes with the fixed weights).

Q6. Drive convention. The companion note's Eq. (20) ($\beta_i = k_{\mathrm{trap},i} L$, from $\delta x = hL/2$) vs. AG13's $F = \frac{1}{2} m \omega_{\mathrm{gw}}^2 \ell_m h$. The current pipeline uses AG13's (full cavity length). Is the L convention in the notes full or half baseline (factor of 2 in the force, 4 in power)?

Q7. FDT temperature. We use the 300 K gas temperature in Eq. (8) with the gas γ_g (feedback treated as noiseless cold damping). Another option to choose could be a temperature scaled by the center of mass.

Q8. Disc mass. Our geometric estimate 5.8×10^{-10} kg (SiO_2 spacer 14.58 μm + two 110 nm Si caps at $r = 75\,\mu$m) matches PRL 128.

Planned next steps

Frequency-dependent weights $\mathbf{w}(f)$ for readout B (optimal post-processing; removes antiresonances); non-uniform G_i study?; stray charge/trap disorder tolerance budget; detection rate objective over the PBH for a few masses?; an axion scan objective (monochromatic source at unknown frequency). Maybe this is where band coverage, and possibly coupling spread arrays, are natively favored.